DBMS COURSE PROJECT · HCI REPORT

Skyward 8 Golden Rules

Where Ben Shneiderman's Eight Golden Rules of Interface Design are implemented in the Skyward Airline Seat Reservation System — with file and line references and screenshots.



STACK

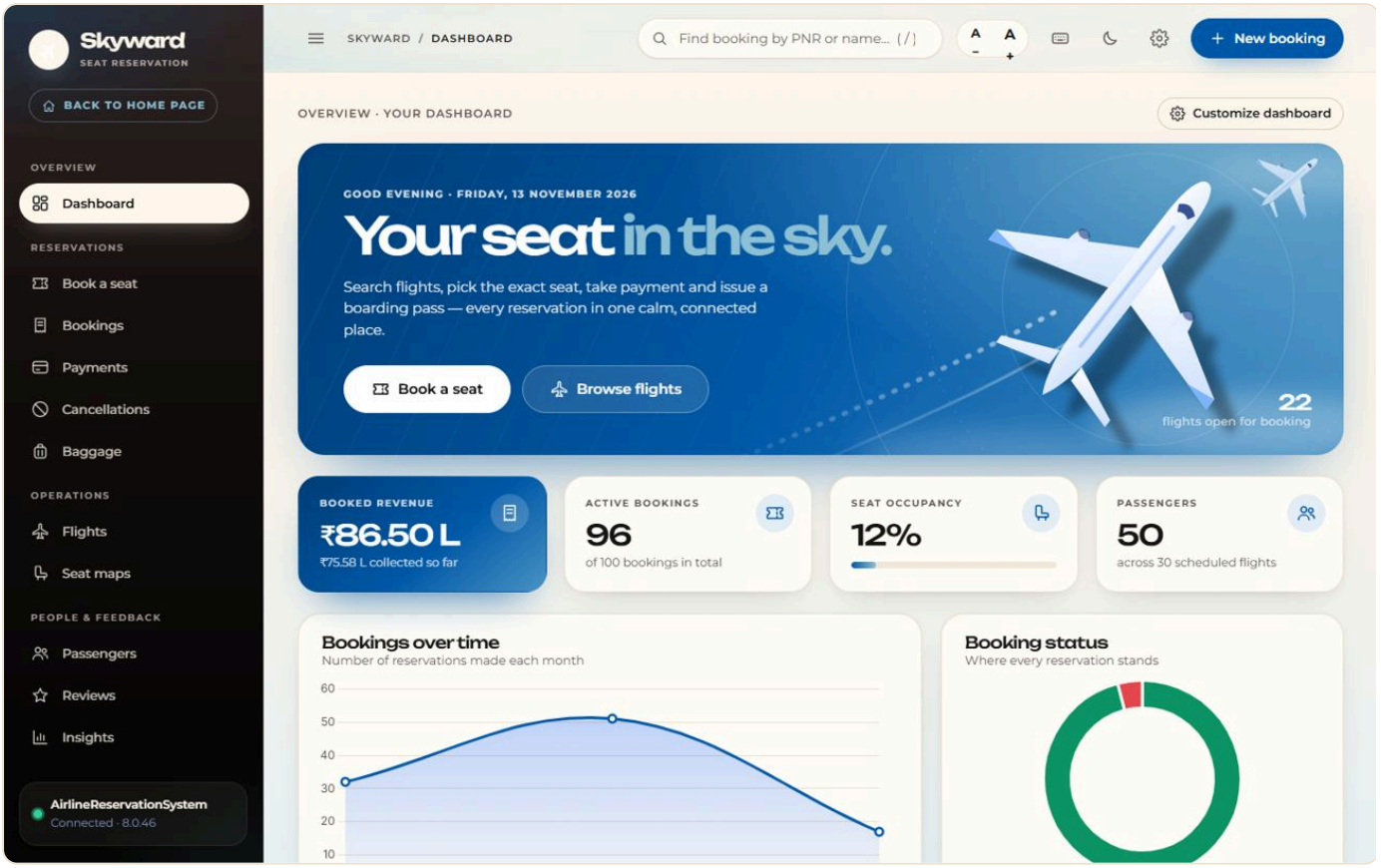
Node.js · Express · MySQL 8 (AirlineReservationSystem) · vanilla JS · GSAP / Lenis / three.js landing page

OVERVIEW

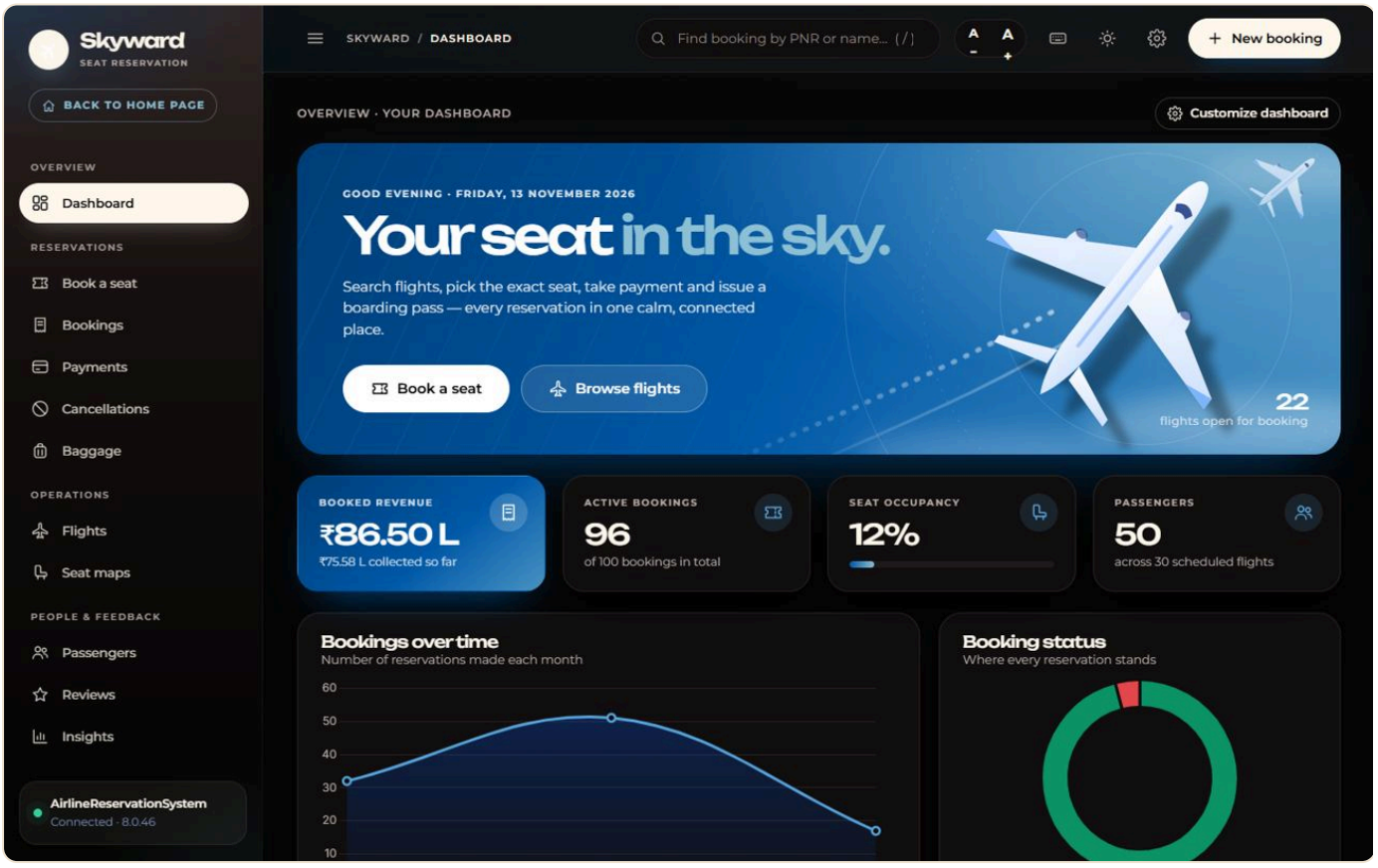
How to read this report

The website has two parts: the **landing page** (`public/index.html`, `css/landing.css`, `js/landing.js`) and the **dashboard** (`public/app.html`). Rule implementations in the dashboard code are tagged with comments `HCI-R1` … `HCI-R8`, so `grep -rn "HCI-R" public` lists every location. File references are relative to the repository root.

#	RULE	HEADLINE IMPLEMENTATION
1	**Strive for Consistency**	One shared design-token set (Unbounded/Montserrat, sky-deep #015AA9, cream #FFF8ED, pill buttons) used by the landing page and every dashboard screen
2	**Seek Universal Usability**	Keyboard shortcuts: g d / g b / g k / g f / g s / g p / g y / g r / g i / g h, n (new booking), / (search), t (theme), + / − (text size), comma (settings), ? (help)
3	**Offer Informative Feedback**	Every create / update / delete / cancel shows a toast with an icon and plain-language message (e.g. 'Flight 6E829 updated — changes saved')
4	**Design Dialogs to Yield Closure**	5-step booking wizard (Flight → Seat → Passenger → Details → Review) with a step indicator
5	**Prevent Errors**	Create / Save stays disabled until every required field is valid; footer says '14 fields still need attention · Show me'
6	**Permit Easy Reversal of Actions**	Delete hides the row immediately and shows '… deleted.' with an UNDO button for 6 s; the DELETE request is only sent when the toast expires
7	**Support Internal Locus of Control**	Settings popover (gear or ',' key): theme, text size, Motion on/off, notifications level, dashboard sections, reset
8	**Reduce Short-Term Memory Load**	Search box shows recent searches and recently viewed bookings (max 6 each, clearable, arrow-key navigation)



Dashboard (light theme) — restyled to match the landing page.



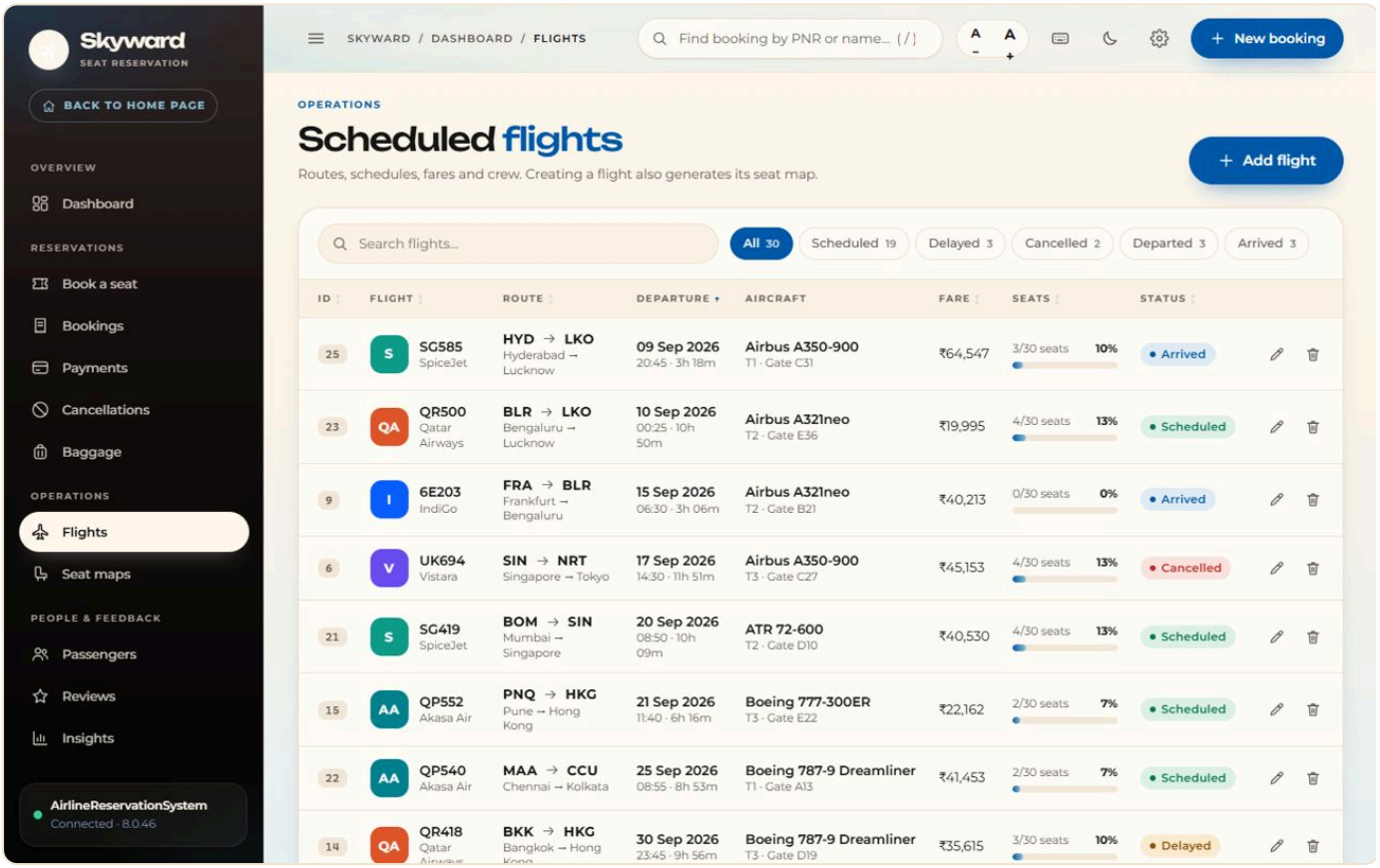
Dashboard (dark theme).

RULE 1 OF 8

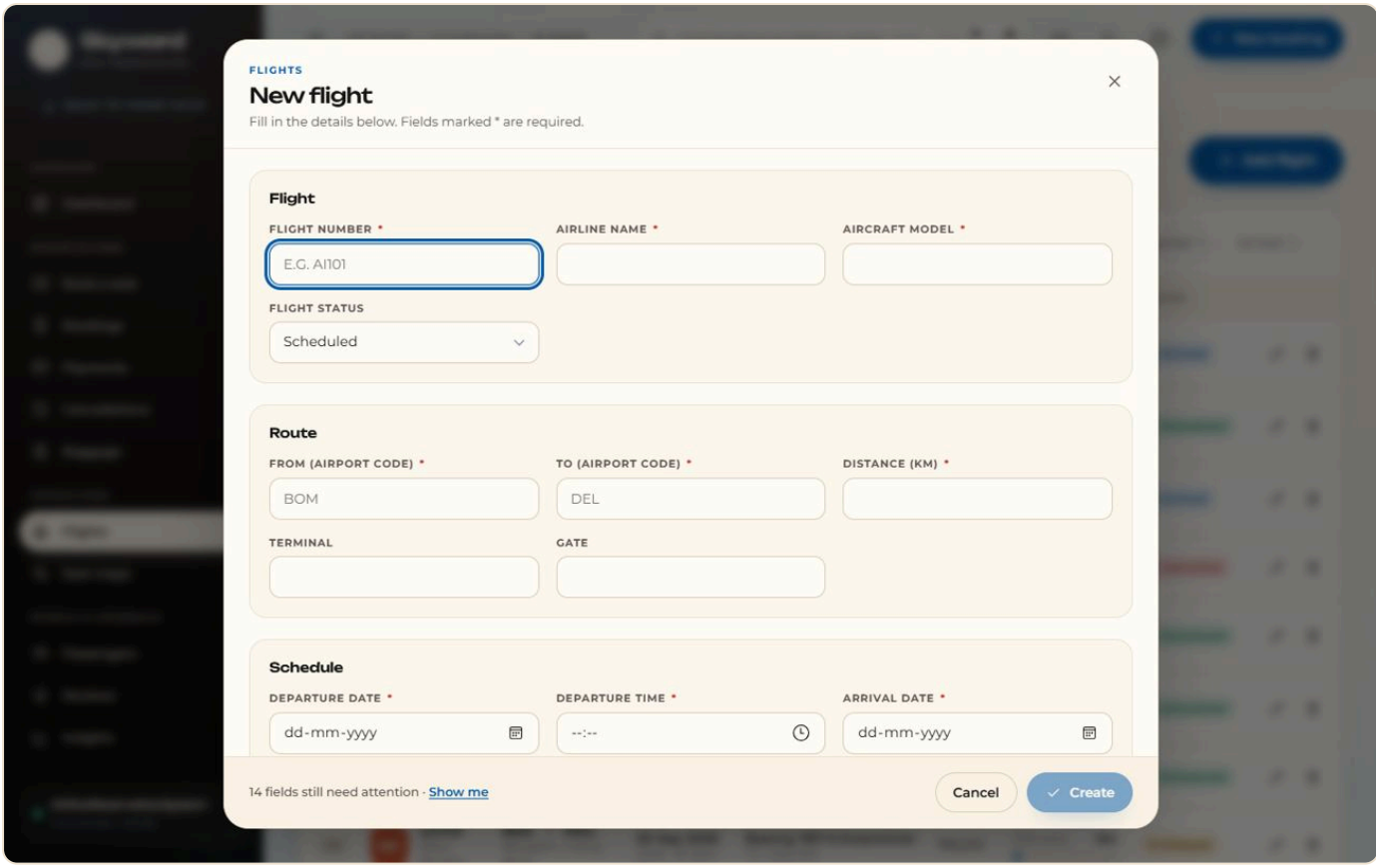
Strive for Consistency

> Use uniform design patterns, terminology, visual styles and user flows across the entire interface.

WHAT WE IMPLEMENTED / HOW THE USER SEES IT	WHERE (FILE:LINE)
One shared design-token set (Unbounded/Montserrat, sky-deep #015AA9, cream #FFF8ED, pill buttons) used by the landing page and every dashboard screen	`public/css/styles.css:22-83, public/css/landing.css`
Every dialog footer is built by one helper: Cancel / secondary on the LEFT, primary on the RIGHT (create, edit, delete, cancel-booking, shortcuts)	`public/js/ui.js:171-175, 554-575; public/js/forms.js:199`
Booking wizard footer is always Back (left) / Continue (right)	`public/js/pages/book.js:245`
Same page header pattern (eyebrow → title → description) and breadcrumb on every page	`public/js/main.js:118`
Shortcut letters shown next to every nav item; brand / 'Back to home page' always leads to the landing page	`public/js/main.js:34; public/app.html:28`
Shared helper styles	`public/css/hci.css:72`



Flights list — same header, filters, table and pill buttons as every other list.



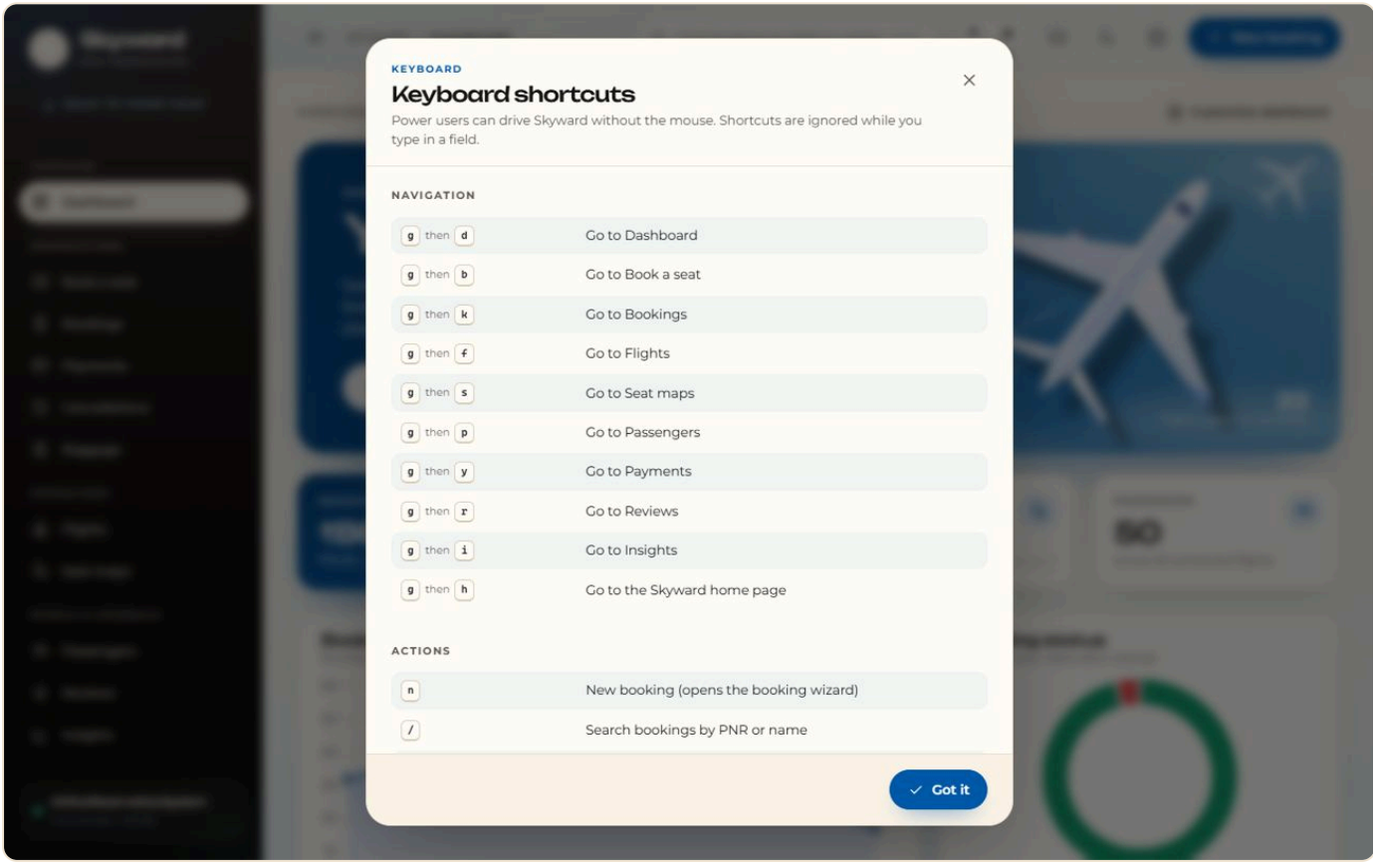
Create dialog — Cancel left, primary right (identical on all forms).

RULE 2 OF 8

Seek Universal Usability

> Design for novices and experts, different abilities and skills — shortcuts for power users, guidance for beginners.

WHAT WE IMPLEMENTED / HOW THE USER SEES IT	WHERE (FILE:LINE)
Keyboard shortcuts: g d / g b / g k / g f / g s / g p / g y / g r / g i / g h, n (new booking), / (search), t (theme), + / − (text size), comma (settings), ? (help)	public/js/hci.js:62-128
'?' opens a Keyboard-shortcuts help dialog listing every shortcut	public/js/hci.js:85
Esc always closes the top-most layer	public/js/ui.js:163
Skip-to-content link (dashboard and landing page)	public/app.html:24; public/js/hci.js:314; public/index.html:31
A− / A+ text-size control, persisted and applied before first paint	public/app.html:14, 51-55; public/js/prefs.js:27, 45
Toasts announced via aria-live; visible focus rings; tables sortable and rows openable with Enter	public/js/ui.js:34; public/css/hci.css:6-70; public/js/pages/crud.js:217-222
Landing page honours prefers-reduced-motion, has keyboard-operable accordion (aria-expanded) and 'D' opens the dashboard	public/js/landing.js:12-39, 282-335; public/index.html:16-28



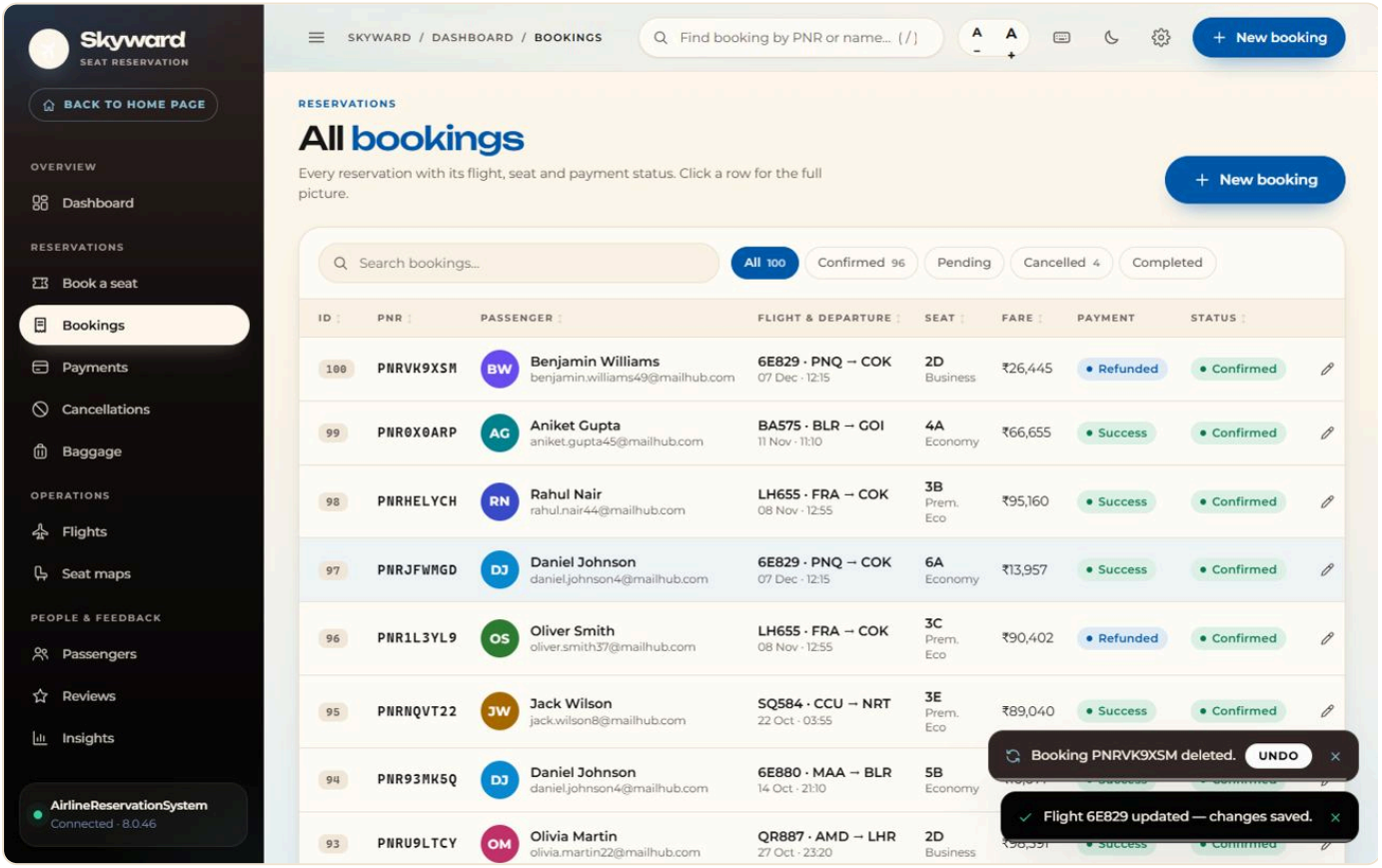
Pressing ? shows every keyboard shortcut.

RULE 3 OF 8

Offer Informative Feedback

> Clear, human-readable responses for every action, scaled to its frequency and importance.

WHAT WE IMPLEMENTED / HOW THE USER SEES IT	WHERE (FILE:LINE)
Every create / update / delete / cancel shows a toast with an icon and plain-language message (e.g. 'Flight 6E829 updated — changes saved')	`public/js/ui.js:34-100`
Thin progress bar at the top while any API request is in flight	`public/js/api.js:7; public/js/hci.js:12`
Buttons show a spinner + 'Working...' while busy (prevents double submit)	`public/js/ui.js:191`
Live database status pill in the sidebar, re-checked every 30 s, with a toast on change	`public/js/main.js:72-74`
Search tells the user how many bookings matched; empty states explain what to do next	`public/js/main.js:52-60`
Landing page shows live figures from MySQL with a 'Live from MySQL · time' status line	`public/js/landing.js:354-394`



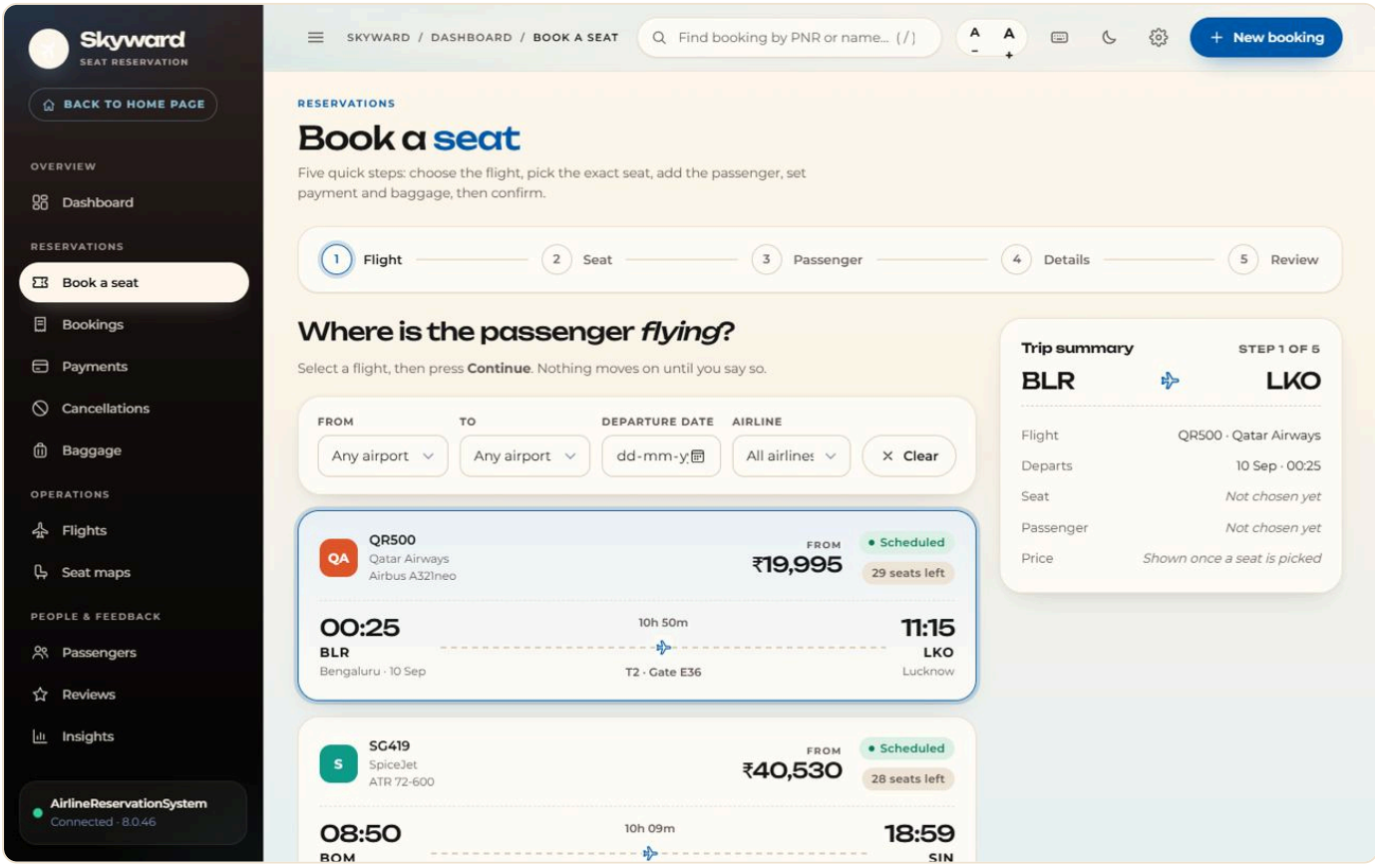
Success toast with icon (bottom right) after saving.

RULE 4 OF 8

Design Dialogs to Yield Closure

> Group actions into sequences with a clear beginning, middle and an explicit completion signal.

WHAT WE IMPLEMENTED / HOW THE USER SEES IT	WHERE (FILE:LINE)
5-step booking wizard (Flight → Seat → Passenger → Details → Review) with a step indicator	`public/js/pages/book.js`
Dedicated 'Booking confirmed' screen: PNR, flight, seat, amount, payment, plus Print boarding pass / Book another / View booking	`public/js/pages/book.js:226-241, 377`
Cancellation completion summary: refund, charge, method, status and 'seat released'	`public/js/forms.js:110-117, 223`
Created / edited / restored rows flash and scroll into view so the user sees the result	`public/js/hci.js:249;` `public/js/forms.js:108, 149;` `public/js/pages/crud.js:178`



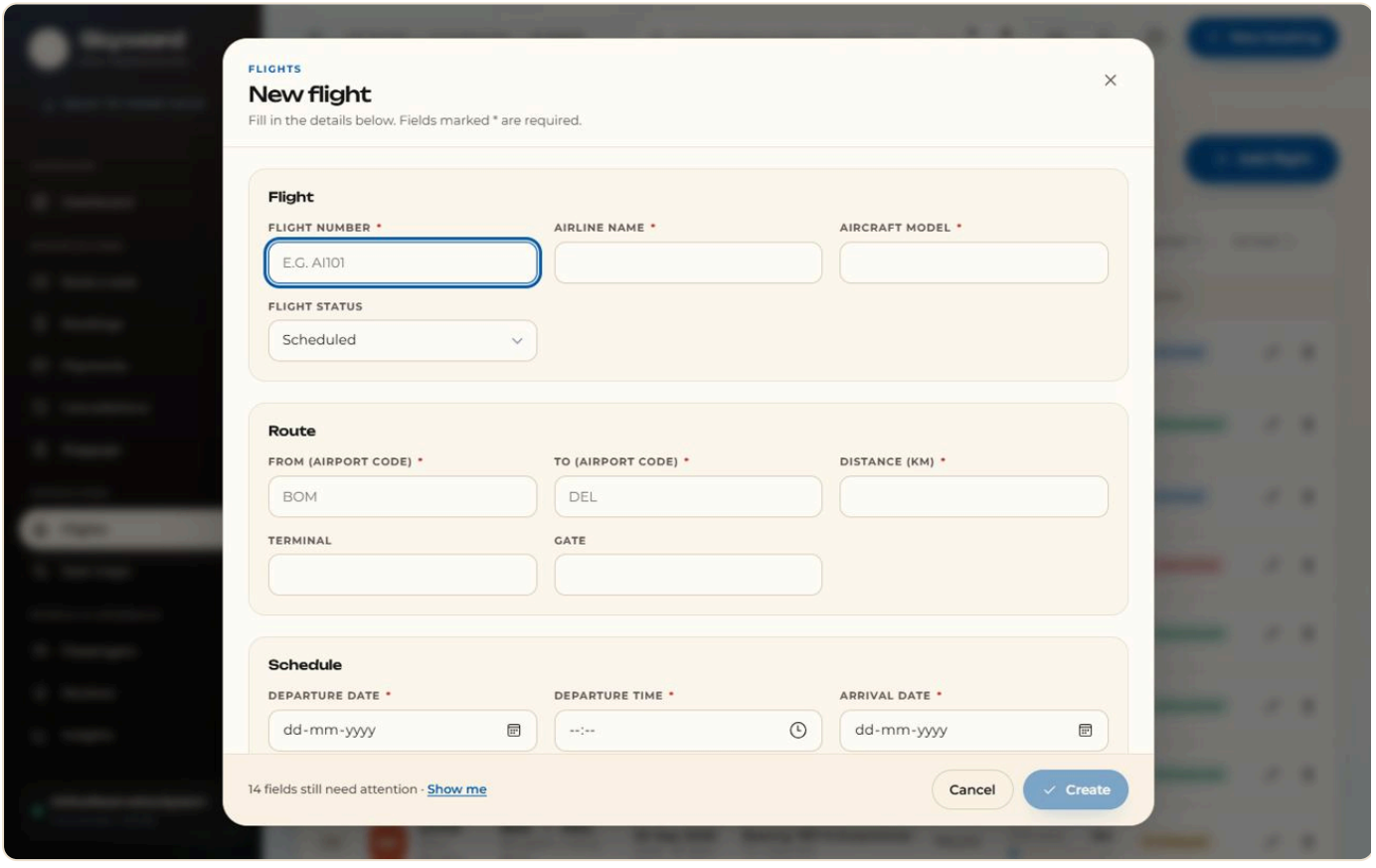
Wizard step 1 of 5 — the step indicator shows the beginning, middle and end of the task.

RULE 5 OF 8

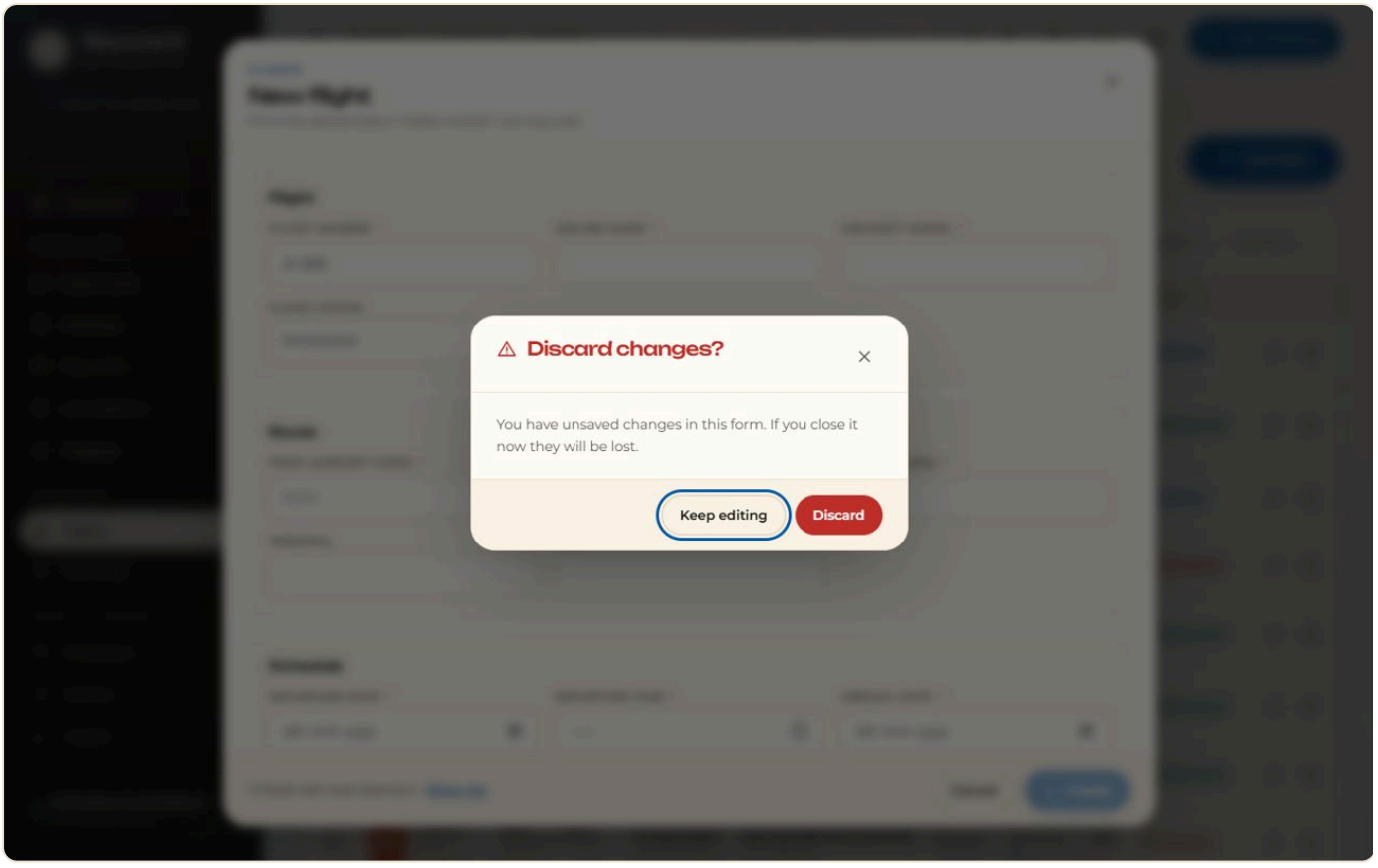
Prevent Errors

> Make serious mistakes hard or impossible: restrict invalid input, validate in real time, use constraints.

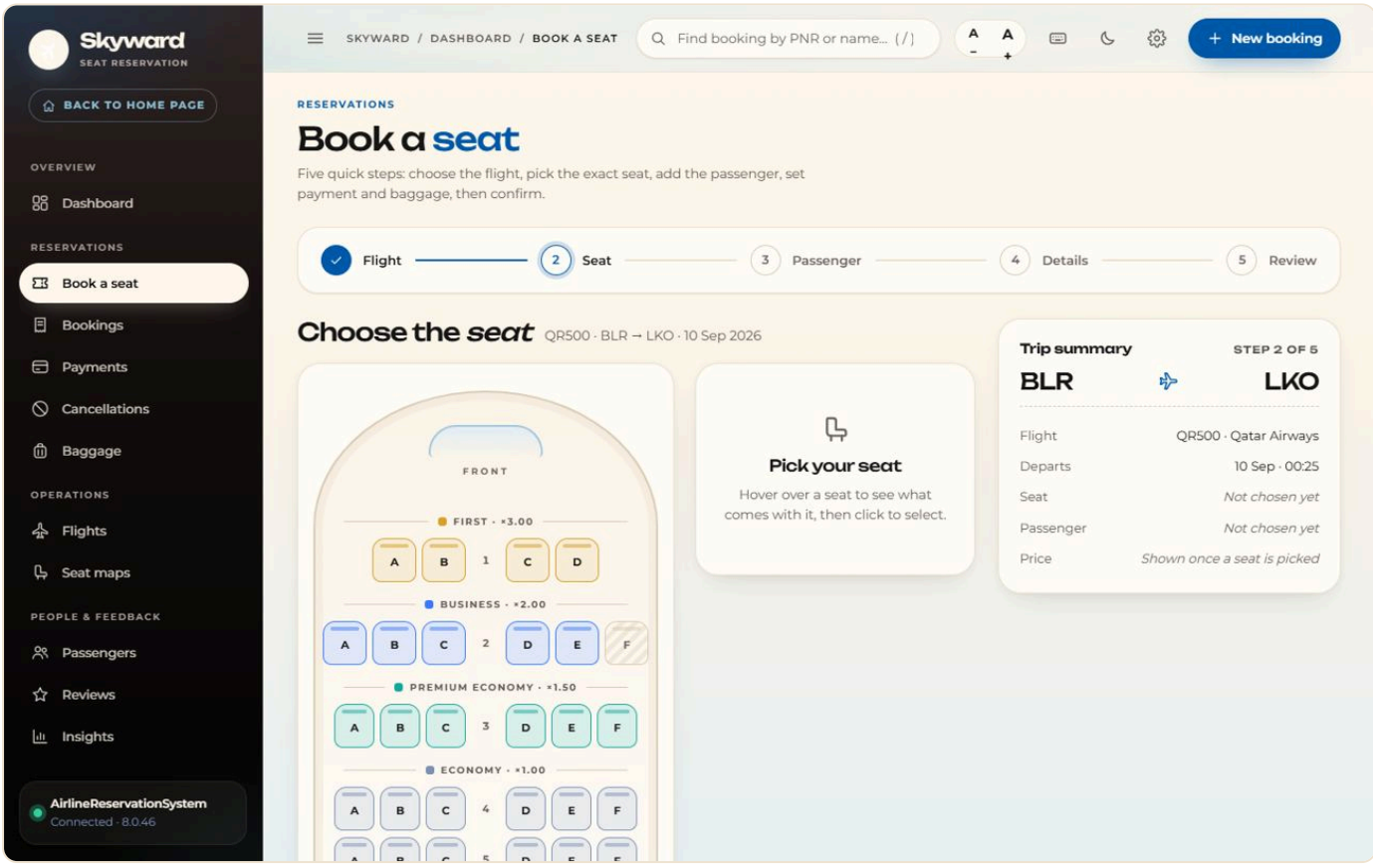
WHAT WE IMPLEMENTED / HOW THE USER SEES IT	WHERE (FILE:LINE)
Create / Save stays disabled until every required field is valid; footer says '14 fields still need attention · Show me'	`public/js/ui.js:554`
Real-time inline validation once a field is visited; one validator used live, on blur and on submit	`public/js/ui.js:318-345, 462`
Date / number constraints: no past departure for new flights, arrival ≥ departure, price / distance / seat counts in range	`public/js/forms.js:44-69;` `public/js/ui.js:391, 448`
Booked / blocked seats are disabled in the seat map	`public/js/seatmap.js:22;` `public/js/pages/book.js:308`
Wizard 'Continue' enabled only when the step is complete	`public/js/pages/book.js:149-160`
Delete always needs confirmation with focus on the safe choice; 'Discard changes?' guard on dirty forms; cancel-booking needs a reason	`public/js/forms.js:164, 183, 202;` `public/js/ui.js:143, 185, 571`



Create is greyed out until the form is valid.



Closing a half-filled form asks 'Discard changes?' (focus on Keep editing).



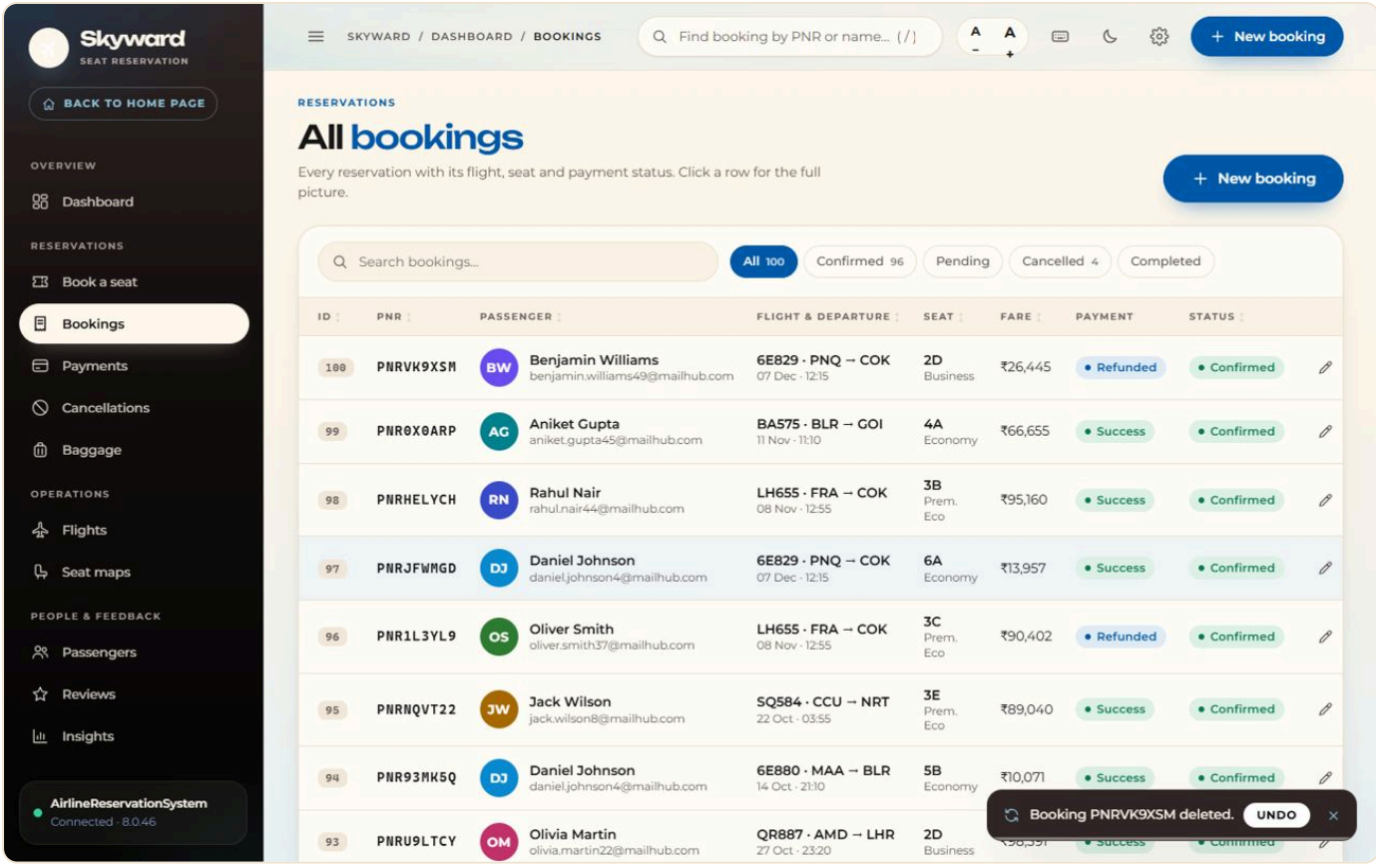
Seat map — booked seats cannot be selected.

RULE 6 OF 8

Permit Easy Reversal of Actions

> Let users undo or cancel actions, reducing anxiety and encouraging exploration.

WHAT WE IMPLEMENTED / HOW THE USER SEES IT	WHERE (FILE:LINE)
Delete hides the row immediately and shows '... deleted.' with an UNDO button for 6 s; the DELETE request is only sent when the toast expires	`public/js/hci.js:208-247;` `public/js/forms.js:170`
Lists and drawers hide rows waiting on Undo; pending deletes flush on route change or when the tab closes	`public/js/pages/crud.js:201;` `public/js/drawers.js:99;` `public/js/main.js:115;` `public/js/hci.js:322`
Coupon apply / remove can be undone	`public/js/pages/book.js:339-352`
Back and the step indicator in the wizard keep every choice already made	`public/js/pages/book.js:255`



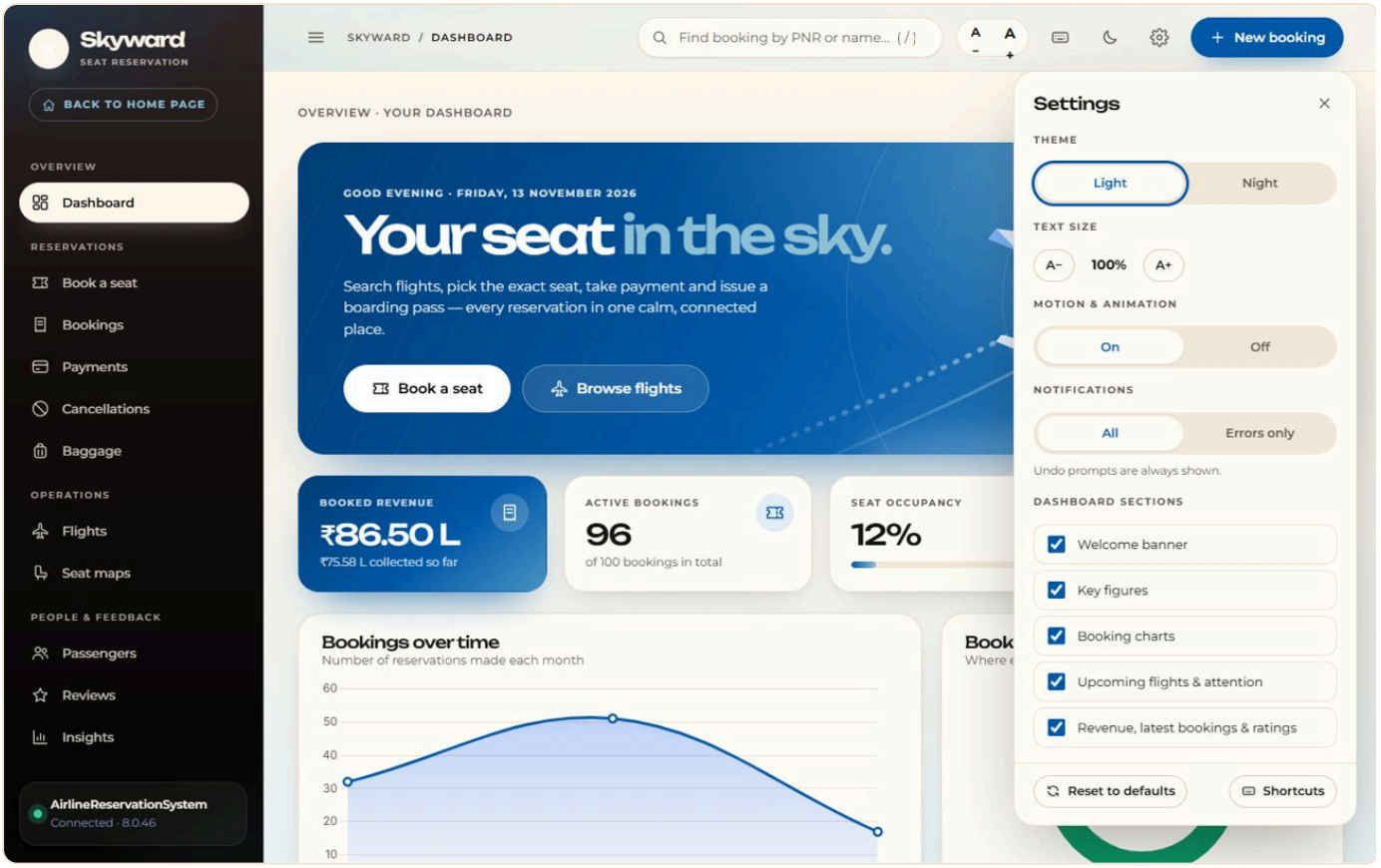
Undo toast after deleting a booking (bottom right).

RULE 7 OF 8

Support Internal Locus of Control

> Users should feel in charge: no surprises, traps or forced automation.

WHAT WE IMPLEMENTED / HOW THE USER SEES IT	WHERE (FILE:LINE)
Settings popover (gear or ',' key): theme, text size, Motion on/off, notifications level, dashboard sections, reset	`public/js/hci.js:129-206; public/app.html:58`
Dashboard is customisable — sections can be shown / hidden live and the choice is remembered	`public/js/pages/dashboard.js:54-56, 110-118`
Every toast can be dismissed (×) and pauses while hovered or focused; 'Errors only' mode	`public/js/ui.js:51, 59, 80`
Motion toggle shared by landing page and dashboard (localStorage 'skyward-motion'); the wizard never auto-advances	`public/js/landing.js:24-32, 221-249; public/js/prefs.js:46; public/js/pages/book.js:258`



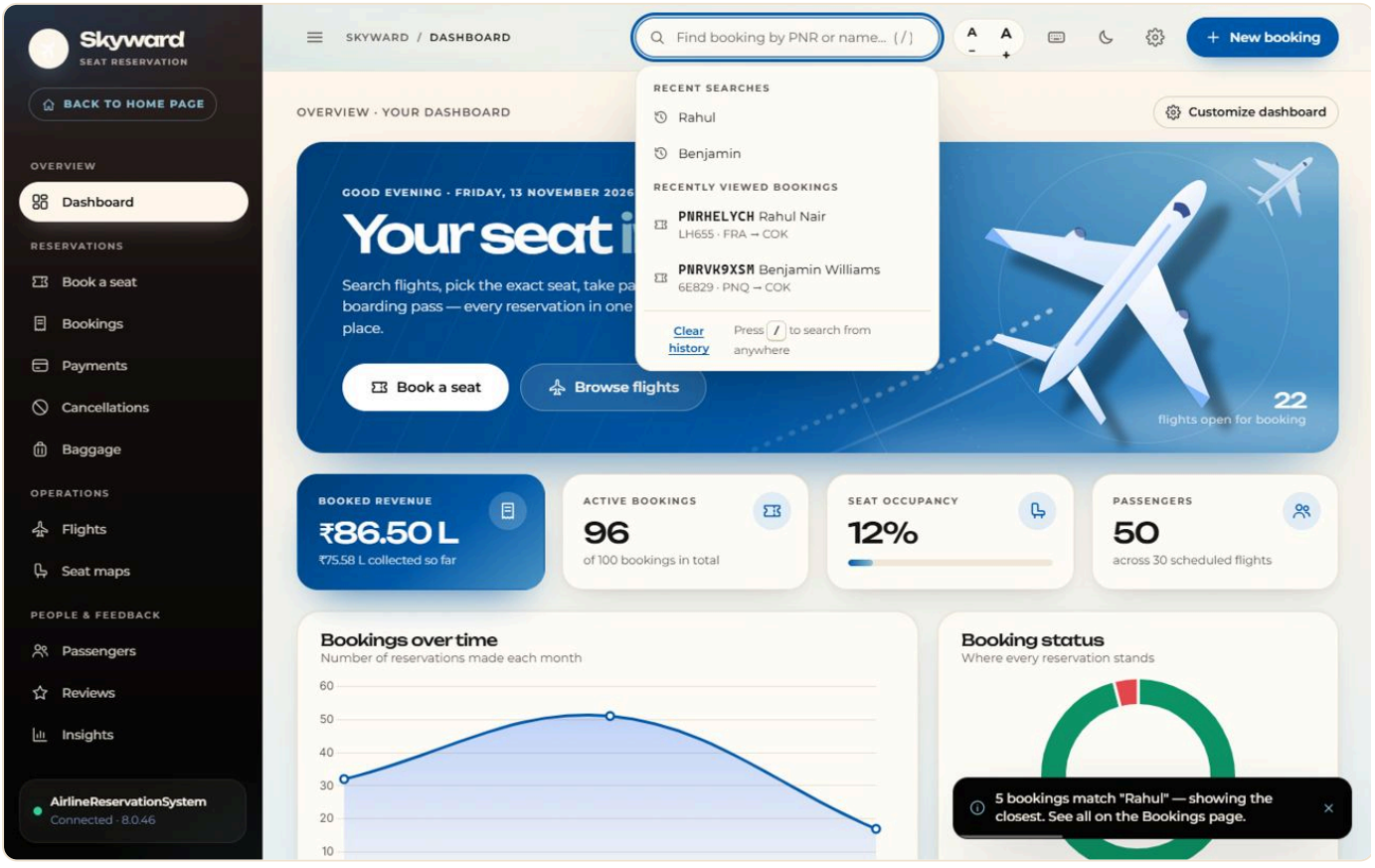
Settings popover — the user decides theme, text size, motion, notifications and dashboard layout.

RULE 8 OF 8

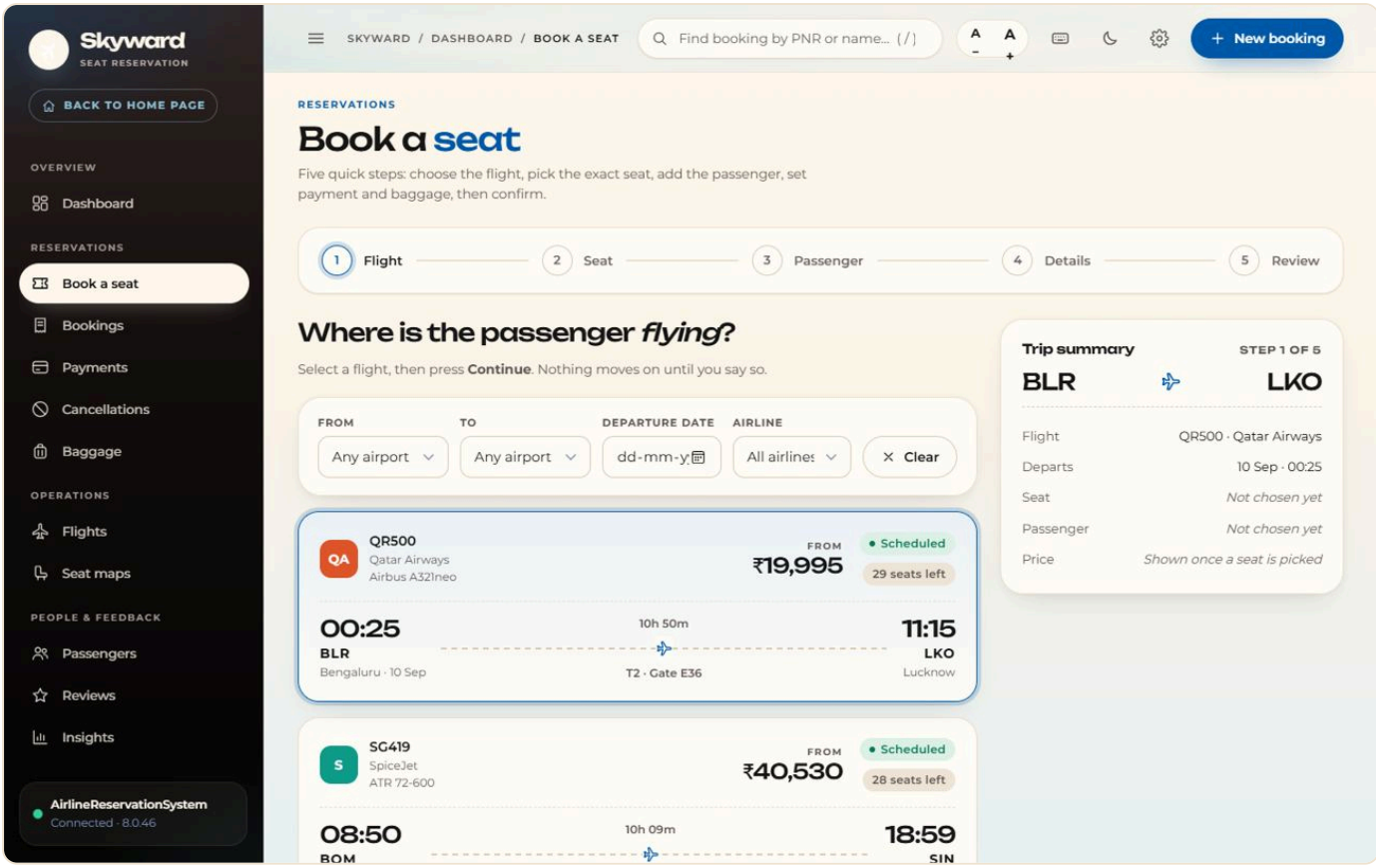
Reduce Short-Term Memory Load

> Rely on recognition over recall: keep options, menus and context visible.

WHAT WE IMPLEMENTED / HOW THE USER SEES IT	WHERE (FILE:LINE)
Search box shows recent searches and recently viewed bookings (max 6 each, clearable, arrow-key navigation)	`public/js/hci.js:259-312`; `public/js/prefs.js:50`; `public/js/drawers.js:100`
Persistent 'Trip summary' in the booking wizard shows flight, seat, passenger and price on every step ('Step N of 5')	`public/js/pages/book.js:57-70`
Breadcrumbs on every page	`public/js/main.js:118`
Each list page remembers its last search / filter / sort with a 'Reset filters' note	`public/js/pages/crud.js:127-139, 162`
Tooltips / labels on all icon buttons	`public/js/drawers.js:95`; `public/app.html:53-59`



Recent searches and recently viewed bookings under the search box.



Trip summary keeps earlier choices visible.